

Arsenii Harbar

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MInf Computer Science & AI, University of Edinburgh | Machine learning at scale, distributed GPU systems & data engineering for financial markets

EDUCATION

University of Edinburgh

Edinburgh, UK

MInf (Integrated Master's) Computer Science & Artificial Intelligence, Year 4

May 2028

- Began university at **16**; **Predicted First Class Honours**. **Year 4**: Machine Learning Systems, Machine Learning at Scale (EPCC, PhD level), Parallel Programming Languages & Systems, Text Technologies for Data Science, Computer Vision (MSc level).
- Self-proposed Honours project**: *Risk-Constrained Reinforcement Learning via Differentiable Convex Optimisation*. A differentiable optimisation layer inside the policy makes hard inventory and cash constraints hold **by construction rather than by penalty**, in a **limit order book simulator**. Scoped and specified independently, then accepted by the school.

RESEARCH & INDUSTRY EXPERIENCE

Research Assistant: Accelerating Multi-Objective Training at Scale

Jun. 2026 – Present

Machine Learning Systems Group, [Dr Luo Mai](#)

Edinburgh, UK

- Research on making **Jacobian computation tractable on GPU clusters**. Training **LLMs** against several objectives at once needs a **per-objective gradient Jacobian** rather than one gradient: the dominant cost of the training step, and why these methods have not scaled. The work targets that bottleneck under **distributed data parallelism** across multiple GPUs and nodes.
- Built the **profiling and benchmarking infrastructure** the group evaluates it with: repeatable campaigns across model sizes, objective counts and batch scales on **SLURM multi-GPU clusters**, with per-phase memory and wall-clock capture and **nsys** kernel attribution. Contributing to a paper in preparation.

1st Place, [NextGenHack](#): NatWest Group × Google Cloud

Mar. 2026 – Jun. 2026

Winning team, then invited presentation at NatWest's Gogarburn headquarters

Edinburgh, UK

- Won first place** as a third-year undergraduate in a predominantly postgraduate field, judged by a **mixed NatWest and Google Cloud panel**. Built the adversarial-robustness component of a multi-signal system for identifying synthetic callers, stress-testing the speech-to-text stage against audio perturbation.
- Selected on that result and **invited to NatWest's Gogarburn headquarters in June 2026 to present the system to the business area that sponsored the event**.

Research Assistant: Data Systems & NLP

Jan. 2025 – Jun. 2026

University of Edinburgh, Prof. Mustapha Douch

Edinburgh, UK

- Built ingestion pipelines over **200M+** unstructured records under tight memory limits, **3×** throughput via vectorised tokenisation and chunked streaming; deployed ensemble classifiers at **97%+** accuracy, **+12 points** over the production baseline, measured before rollout. Largely independent across an **18-month** engagement.

LEADERSHIP

Head of Machine Learning

Sep. 2024 – Present

Algorithmic Trading & Quantitative Finance Society (QuantSoc), University of Edinburgh

Edinburgh, UK

- Corporate sponsors and partners**: [Hudson River Trading](#), [Jane Street](#) and [TPP](#). **Personally secured Jane Street and TPP**, from cold outreach through to signed partnership.
- Lead the machine learning, optimisation and algorithms workstream for a society of **500+** active members: **120+** strategies researched, **50,000+** backtests executed on university GPU compute with experiment scheduling and result capture at scale.
- Deliver **live workshops on signal generation and execution algorithms to 100+** students; built and operate the society's technical platform end to end, including timed competitions and a quant interview bank with **server-verified rankings**.
- Negotiated **inter-university chapter partnerships** with **Warwick, Glasgow, TU Dublin, TU Munich and Hong Kong**, taking the society from one campus to an international structure. Two published technical articles on [quant-soc.com](#).

PROJECTS

RiskBeacon: Real-Time Market Risk Monitoring | [Python](#), [FastAPI](#), [FinBERT](#), [Docker](#), [MLflow](#) | [GitHub](#)

2025

- Ingests OHLCV market data and computes microstructure and stress metrics, including **VPIN**, **Yang-Zhang volatility** and **HMM market-regime classification**, at a **measured median ingest latency of 14.16 ms**. FinBERT sentiment service, Interactive Brokers feed, **207 tests** with mutation and property-based testing, CI. **Awarded full marks** on Informatics Large Practical.

AWARDS & CERTIFICATIONS

- JPMorganChase Code for Good**, Glasgow 2025. **1st place**, Skyscanner × Google AccessAlthon (2026) and [AI Expo](#), Edinburgh AI. **1 of 200 builders from 1,500+ applicants**, EuroTech Federation, TU Munich. **Certifications**: [Deep Learning Specialization](#) (DeepLearning.AI), [GPU Programming](#) (Johns Hopkins).

TECHNICAL SKILLS

Python, SQL, C/C++ (incl. CUDA), Java, Haskell | PyTorch, HuggingFace, distributed data-parallel training, PPO & multi-objective RL, MLflow | SLURM multi-GPU clusters, **nsys** profiling, Docker, CI/CD